

Week 5 — Optimization

Unconstrained Optimization: Global and Local Maxima

Let $X \subseteq \mathbb{R}^J$ and let

$$f : X \rightarrow \mathbb{R}.$$

We are interested in choosing $x^* \in X$ in order to maximize f .

Definition (Global maximizer). A point $x^* \in X$ is a *global maximizer* of f on X if

$$f(x^*) \geq f(x) \quad \text{for every } x \in X.$$

Equivalently,

$$x^* \in \arg \max_{x \in X} f(x).$$

Definition (Local maximizer). Let (X, d) be a metric space. A point $x^* \in X$ is a *local maximizer* of f if there exists $\varepsilon > 0$ such that

$$f(x^*) \geq f(x)$$

for every $x \in X$ satisfying

$$d(x, x^*) < \varepsilon.$$

Equivalently, x^* maximizes f on some neighborhood of x^* relative to the feasible set X .

Remark. Every global maximizer is automatically a local maximizer. The converse is false in general: a function may have several local maxima, only some of which are global.

Critical Points and First-Order Necessary Conditions

Suppose now that $X \subseteq \mathbb{R}^J$ and f is differentiable.

Definition (Critical point). A point $x^* \in X$ is a *critical point* of f if

$$\nabla f(x^*) = 0,$$

where

$$\nabla f(x^*) = \begin{pmatrix} \frac{\partial f}{\partial x_1}(x^*) \\ \vdots \\ \frac{\partial f}{\partial x_J}(x^*) \end{pmatrix}.$$

Lemma (Fermat's necessary condition). Let $f : X \rightarrow \mathbb{R}$ be differentiable. If

$$x^* \in \text{int}(X)$$

is a local maximizer of f , then

$$\nabla f(x^*) = 0.$$

Thus every interior local maximizer is a critical point.

Intuition. If x^* is an interior point, we can move a sufficiently small distance from x^* in every coordinate direction. If some partial derivative were strictly positive or strictly negative, there would be a nearby direction in which f increases, contradicting local maximality.

Remark (Why interiority matters). A boundary maximizer need not satisfy the first-order condition $\nabla f(x^*) = 0$.

Thus:

- an interior local maximizer must be a critical point;
- a boundary local maximizer need not be a critical point;
- a critical point need not be a local maximizer.

For example, an interior local minimum also satisfies

$$\nabla f(x^*) = 0,$$

even though it is not a local maximum.

Why Concavity Is Powerful for Maximization

Concavity greatly strengthens local optimality conditions.

Proposition (Local and global maxima). *Let $X \subseteq \mathbb{R}^J$ be convex and let $f : X \rightarrow \mathbb{R}$. If f is either concave or strictly quasiconcave, then*

$$x^* \text{ is a global maximizer} \iff x^* \text{ is a local maximizer.}$$

Proof. The direction

$$\text{global maximizer} \implies \text{local maximizer}$$

is immediate.

For the converse, we prove the contrapositive. Suppose x^* is not a global maximizer. Then there exists $y \in X$ such that

$$f(y) > f(x^*).$$

For $\alpha \in (0, 1)$ define

$$x(\alpha) = \alpha y + (1 - \alpha)x^*.$$

Since X is convex,

$$x(\alpha) \in X.$$

Moreover,

$$x(\alpha) \rightarrow x^* \quad \text{as } \alpha \downarrow 0.$$

If f is strictly quasiconcave, then

$$f(x(\alpha)) > \min\{f(y), f(x^*)\} = f(x^*).$$

If f is concave, then

$$f(x(\alpha)) \geq \alpha f(y) + (1 - \alpha)f(x^*) > f(x^*).$$

Thus, arbitrarily close to x^* , there are feasible points with strictly larger objective value. Hence x^* cannot be a local maximizer. $\square$

Intuition. Under concavity or strict quasiconcavity, if there exists any point y that is better than x^* , then every sufficiently small movement from x^* toward y already improves the objective. Therefore a “bad local maximum” cannot exist.

Proposition (Critical points under concavity). *Let $X \subseteq \mathbb{R}^J$ be convex, let $f : X \rightarrow \mathbb{R}$ be differentiable and concave, and let*

$$x^* \in \text{int}(X).$$

Then

$$x^* \text{ is a global maximizer} \iff \nabla f(x^*) = 0.$$

Consequently,

$$x^* \text{ is a local maximizer} \iff \nabla f(x^*) = 0.$$

Remark (Important qualification). Strict quasiconcavity alone is enough to make every local maximizer global, but it is *not* enough to make every critical point a maximizer.

For example,

$$f(x) = x^3$$

is strictly increasing, hence strictly quasiconcave on $\mathbb{R}$, and

$$f'(0) = 0,$$

but 0 is not a local maximizer.

Thus the critical-point characterization above should be associated with concavity, not with strict quasiconcavity alone.

Intuition. This is one of the main reasons concavity is so useful in optimization. Once concavity is known, an interior first-order condition is not merely necessary: it is sufficient for global optimality. Hence we often do not need to inspect second-order conditions separately.

Second-Order Conditions

Let $f : X \rightarrow \mathbb{R}$ be twice continuously differentiable and let

$$H_f(x)$$

denote its Hessian matrix.

Proposition (Second-order necessary condition). *Suppose*

$$x^* \in \text{int}(X)$$

is a local maximizer of f . Then

$$\nabla f(x^*) = 0$$

and

$$H_f(x^*)$$

is negative semidefinite:

$$v^\top H_f(x^*) v \leq 0 \quad \text{for every } v \in \mathbb{R}^J.$$

Proposition (Second-order sufficient condition). *Let $X \subseteq \mathbb{R}^J$ be open and let $f : X \rightarrow \mathbb{R}$ be twice continuously differentiable.*

If

$$\nabla f(x^*) = 0$$

and

$$H_f(x^*)$$

is negative definite, i.e.

$$v^\top H_f(x^*)v < 0 \quad \text{for every } v \neq 0,$$

then x^* is a strict local maximizer of f .

Remark. The distinction between negative semidefinite and negative definite matters.

A local maximum only implies

$$H_f(x^*) \preceq 0.$$

Negative definiteness,

$$H_f(x^*) \prec 0,$$

is a stronger condition and provides a sufficient condition for a strict local maximum.

Equality-Constrained Optimization

Let

$$X \subseteq \mathbb{R}^J$$

be open and let

$$f : X \rightarrow \mathbb{R}$$

be the objective function.

Suppose there are M equality constraints

$$g_m : X \rightarrow \mathbb{R}, \quad m = 1, \dots, M.$$

The standard equality-constrained maximization problem is

$$\max_{x \in X} f(x)$$

subject to

$$g_m(x) = 0, \quad m = 1, \dots, M.$$

The feasible set is

$$C = \{x \in X : g_m(x) = 0, \ m = 1, \dots, M\}.$$

Remark (When the original domain is not open). Many economic problems are naturally defined on non-open domains such as

$$\mathbb{R}_+^J.$$

There are two common ways to proceed.

1. First analyze the problem on

$$\text{int}(X),$$

and then separately check whether an optimum may occur on the boundary

$$X \setminus \text{int}(X).$$

2. Define the objective on a larger open set, such as

$$\mathbb{R}^J,$$

and impose the original domain restrictions explicitly as inequality constraints, for example

$$x_i \geq 0.$$

Inequality constraints require additional tools, which will be treated separately.

Remark (Putting equality constraints in standard form). A constraint of the form

$$g_m(x) = b_m$$

can always be rewritten as

$$\tilde{g}_m(x) = 0,$$

where

$$\tilde{g}_m(x) := g_m(x) - b_m.$$

Thus there is no loss of generality in writing equality constraints in the standard form

$$g_m(x) = 0.$$

A Two-Dimensional Equality-Constrained Problem

Consider

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}$$

and the problem

$$\max_{(x_1, x_2) \in \mathbb{R}^2} f(x_1, x_2)$$

subject to

$$g(x_1, x_2) = 0,$$

where, in the class example,

$$g(x_1, x_2) = x_1^2 + x_2^2 - 1.$$

Thus the feasible set is the unit circle:

$$x_1^2 + x_2^2 = 1.$$

Suppose f is strictly increasing and strictly concave. Its level curves have the usual downward-sloping, convex-to-the-origin shape.

At an interior smooth solution

$$x^* = (x_1^*, x_2^*),$$

the highest attainable level curve of f is tangent to the constraint

$$g(x_1, x_2) = 0.$$

Intuition. If the objective level curve crossed the constraint transversally at x^* , we could move a small distance along the constraint and reach a higher level of f . At an optimum, the two curves therefore have the same tangent direction.

Implicit Function Theorem and the Slope of a Level Curve

Let

$$k^* = f(x_1^*, x_2^*).$$

Suppose

$$\frac{\partial f}{\partial x_2}(x_1^*, x_2^*) \neq 0.$$

By the Implicit Function Theorem, there exists a continuously differentiable function

$$\hat{f}$$

defined on an interval around x_1^* such that

$$f(x_1, \hat{f}(x_1)) = k^*,$$

and

$$\hat{f}(x_1^*) = x_2^*.$$

Differentiating

$$f(x_1, \hat{f}(x_1)) = k^*$$

with respect to x_1 gives

$$\frac{\partial f}{\partial x_1} + \frac{\partial f}{\partial x_2} \hat{f}'(x_1) = 0.$$

Hence, at x^* ,

$$\hat{f}'(x_1^*) = -\frac{\frac{\partial f}{\partial x_1}(x^*)}{\frac{\partial f}{\partial x_2}(x^*)}.$$

Thus the slope of the objective level curve at x^* is

$$-\frac{f_{x_1}(x^*)}{f_{x_2}(x^*)}.$$

Implicit Function Theorem and the Constraint Curve

Similarly, suppose

$$\frac{\partial g}{\partial x_2}(x^*) \neq 0.$$

Then the Implicit Function Theorem gives a continuously differentiable function

$$\hat{g}$$

defined near x_1^* such that

$$g(x_1, \hat{g}(x_1)) = 0,$$

and

$$\hat{g}(x_1^*) = x_2^*.$$

Differentiating yields

$$g_{x_1} + g_{x_2} \hat{g}'(x_1) = 0,$$

so

$$\boxed{\widehat{g}'(x_1^*) = -\frac{g_{x_1}(x^*)}{g_{x_2}(x^*)}}.$$

Hence the slope of the constraint curve at x^* is

$$-\frac{g_{x_1}(x^*)}{g_{x_2}(x^*)}.$$

Tangency Condition

At an interior smooth constrained optimum, the objective level curve and the constraint curve are tangent. Therefore their slopes coincide:

$$-\frac{f_{x_1}(x^*)}{f_{x_2}(x^*)} = -\frac{g_{x_1}(x^*)}{g_{x_2}(x^*)}.$$

Equivalently,

$$\frac{f_{x_1}(x^*)}{f_{x_2}(x^*)} = \frac{g_{x_1}(x^*)}{g_{x_2}(x^*)}.$$

Cross-multiplying,

$$f_{x_1}(x^*)g_{x_2}(x^*) = f_{x_2}(x^*)g_{x_1}(x^*).$$

Hence the two gradient vectors are proportional: there exists a scalar λ^* such that

$$\boxed{\nabla f(x^*) = \lambda^* \nabla g(x^*)}.$$

In coordinates,

$$\begin{aligned} f_{x_1}(x^*) - \lambda^* g_{x_1}(x^*) &= 0, \\ f_{x_2}(x^*) - \lambda^* g_{x_2}(x^*) &= 0. \end{aligned}$$

These are two equations in the three unknowns

$$x_1^*, \quad x_2^*, \quad \lambda^*.$$

The third equation is feasibility:

$$g(x_1^*, x_2^*) = 0.$$

Thus the candidate solution is characterized by

$$\boxed{\begin{cases} f_{x_1}(x^*) - \lambda^* g_{x_1}(x^*) = 0, \\ f_{x_2}(x^*) - \lambda^* g_{x_2}(x^*) = 0, \\ g(x^*) = 0. \end{cases}}$$

The Lagrangian

The previous equations can be collected into a single auxiliary function.

Definition (Lagrangian). For the equality-constrained problem

$$\max_x f(x) \quad \text{subject to } g(x) = 0,$$

define the *Lagrangian*

$$\mathcal{L}(x, \lambda) = f(x) - \lambda g(x).$$

In the two-dimensional case,

$$\mathcal{L}(x_1, x_2, \lambda) = f(x_1, x_2) - \lambda g(x_1, x_2).$$

The critical-point conditions are

$$\frac{\partial \mathcal{L}}{\partial x_1} = f_{x_1}(x) - \lambda g_{x_1}(x) = 0,$$

$$\frac{\partial \mathcal{L}}{\partial x_2} = f_{x_2}(x) - \lambda g_{x_2}(x) = 0,$$

and

$$\frac{\partial \mathcal{L}}{\partial \lambda} = -g(x) = 0.$$

Therefore,

$$\nabla_{(x, \lambda)} \mathcal{L}(x^*, \lambda^*) = 0$$

reproduces exactly the tangency conditions together with feasibility.

Remark (Sign convention). One may equivalently define

$$\mathcal{L}(x, \lambda) = f(x) + \lambda g(x).$$

This merely replaces the multiplier λ by $-\lambda$. For equality constraints the sign convention has no substantive effect.

Geometric Interpretation of the Lagrange Condition

The Lagrange condition can also be understood directly from the geometry of level curves, tangent directions, and gradients.

Consider the equality-constrained problem

$$\max_{x \in \mathbb{R}^2} f(x) \quad \text{subject to } g(x) = 0.$$

Suppose that x^* is a regular constrained optimum, meaning in particular that

$$\nabla g(x^*) \neq 0.$$

Let

$$k^* := f(x^*).$$

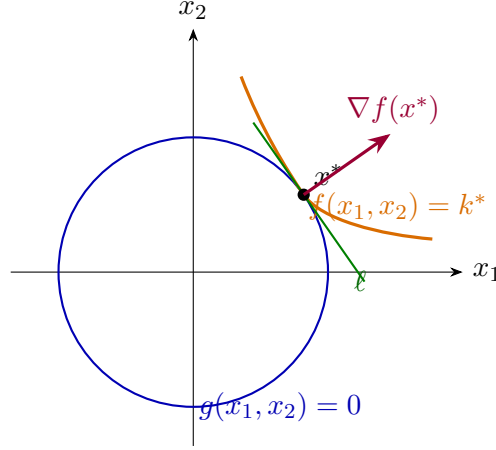


Figure 1: At a smooth constrained optimum, the objective level curve $f(x) = k^*$ is tangent to the constraint $g(x) = 0$. The gradient $\nabla f(x^*)$ is normal to the objective level curve.

Then x^* belongs simultaneously to the objective level curve

$$\{x : f(x) = k^*\}$$

and to the constraint

$$\{x : g(x) = 0\}.$$

At a smooth constrained optimum, these two curves are tangent.

Intuition. The picture contains the essential geometry behind the Lagrange condition. The orange curve represents combinations of (x_1, x_2) that give the same objective value k^* . The blue curve is the feasible set. At x^* , the highest attainable objective level curve is tangent to the constraint.

The common tangent line is denoted by ℓ . The key question is therefore: what is the relationship between the gradient of a function and the tangent line to one of its level curves?

Directional Derivatives and Level Curves

Definition (Directional derivative). Let $f : \mathbb{R}^J \rightarrow \mathbb{R}$ be differentiable and let $v \in \mathbb{R}^J$ be a direction vector. The directional derivative of f at x in the direction v is

$$D_v f(x) = \nabla f(x)^\top v.$$

If

$$\|v\| = 1,$$

then $D_v f(x)$ measures the instantaneous rate of change of f per unit distance in the direction v .

Now suppose that v is a tangent vector to the level curve

$$f(x) = k^*$$

at x^* .

More formally, suppose there exists a differentiable curve

$$\gamma : (-\varepsilon, \varepsilon) \rightarrow \mathbb{R}^J$$

such that

$$\gamma(0) = x^*, \quad \gamma'(0) = v,$$

and

$$f(\gamma(t)) = k^*$$

for all t sufficiently close to zero.

Because f remains constant as we move along the level curve,

$$\left. \frac{d}{dt} f(\gamma(t)) \right|_{t=0} = 0.$$

By the chain rule,

$$\nabla f(x^*)^\top \gamma'(0) = 0.$$

Since $\gamma'(0) = v$,

$$D_v f(x^*) = \nabla f(x^*)^\top v = 0.$$

Therefore,

$$\boxed{\nabla f(x^*) \perp v}.$$

Since v points along the tangent line ℓ , this means

$$\boxed{\nabla f(x^*) \perp \text{the level curve of } f}.$$

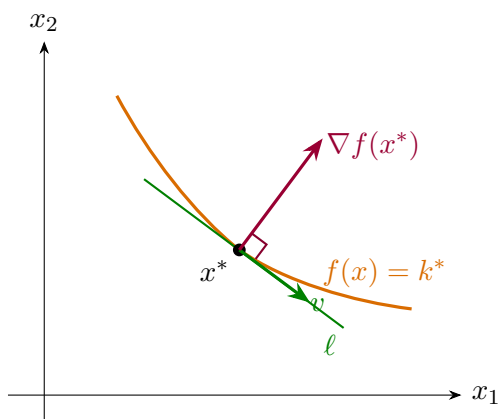


Figure 2: A tangent direction v produces no first-order change in f . Hence $\nabla f(x^*)^\top v = 0$, so the gradient is perpendicular to the level curve.

Intuition. The gradient points in the direction of steepest increase of the function. A tangent direction to a level curve, by contrast, is a direction in which the function does not change to first order.

These two directions must therefore be orthogonal.

Applying the Same Argument to the Constraint

Exactly the same argument applies to the constraint curve

$$g(x) = 0.$$

If v is tangent to the constraint at x^* , then moving infinitesimally in direction v keeps us on the constraint to first order. Therefore,

$$D_v g(x^*) = 0.$$

Hence

$$\nabla g(x^*)^\top v = 0,$$

and thus

$$\boxed{\nabla g(x^*) \perp \text{the constraint curve}}.$$

At a constrained optimum, the objective level curve and the constraint have the same tangent line. Therefore

$$\nabla f(x^*)$$

and

$$\nabla g(x^*)$$

are both perpendicular to the same tangent space.

In $\mathbb{R}^2$, this implies that they are parallel. Hence there exists a scalar $\lambda^* \in \mathbb{R}$ such that

$$\boxed{\nabla f(x^*) = \lambda^* \nabla g(x^*)}.$$

This is precisely the Lagrange first-order condition.

Geometric Meaning of the Multiplier's Sign

The gradient $\nabla g(x^*)$ determines a normal direction to the constraint, but the same normal line has two possible orientations:

$$\nabla g(x^*) \quad \text{and} \quad -\nabla g(x^*).$$

The multiplier λ^* determines which orientation of this normal matches $\nabla f(x^*)$.

Remark (The sign of the multiplier). For an equality constraint, the multiplier is unrestricted in sign:

$$\lambda^* \in \mathbb{R}.$$

Under the convention

$$\mathcal{L}(x, \lambda) = f(x) - \lambda g(x),$$

the first-order condition is

$$\nabla f(x^*) = \lambda^* \nabla g(x^*).$$

Therefore:

$$\lambda^* > 0$$

when $\nabla f(x^*)$ and $\nabla g(x^*)$ point in the same direction, whereas

$$\lambda^* < 0$$

when they point in opposite directions.

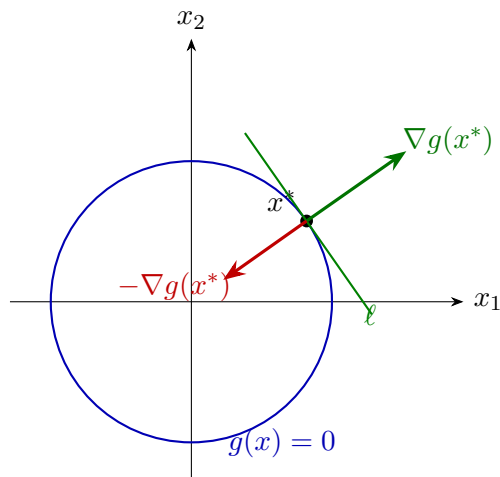


Figure 3: The gradient of the constraint is normal to the feasible curve. Changing the sign of the constraint reverses the normal direction.

However, this sign is not intrinsic to the economic problem. Replacing

$$g(x) = 0$$

by the equivalent constraint

$$-g(x) = 0$$

reverses the direction of the constraint gradient and changes the sign of λ^* .

The Unit-Circle Example Revisited

Consider

$$g(x_1, x_2) = x_1^2 + x_2^2 - 1.$$

The feasible set is

$$g(x_1, x_2) = 0 \iff x_1^2 + x_2^2 = 1.$$

The gradient is

$$\nabla g(x_1, x_2) = \begin{pmatrix} 2x_1 \\ 2x_2 \end{pmatrix}.$$

At any point on the unit circle,

$$x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix},$$

we therefore have

$$\nabla g(x) = 2x.$$

Hence $\nabla g(x)$ points radially outward from the origin and is normal to the circle.

At a regular constrained optimum x^* ,

$$\nabla f(x^*) = \lambda^* \nabla g(x^*),$$

so

$$\nabla f(x^*) = \lambda^* \begin{pmatrix} 2x_1^* \\ 2x_2^* \end{pmatrix}.$$

Equivalently,

$$\begin{cases} f_{x_1}(x^*) - 2\lambda^* x_1^* = 0, \\ f_{x_2}(x^*) - 2\lambda^* x_2^* = 0, \\ (x_1^*)^2 + (x_2^*)^2 = 1. \end{cases}$$

The first two equations impose the geometric condition that

$$\nabla f(x^*) \parallel \nabla g(x^*),$$

while the third equation imposes feasibility.

Thus the three equations determine the candidate values

$$x_1^*, \quad x_2^*, \quad \lambda^*.$$

Intuition. The constraint leaves only one locally feasible direction: the direction tangent to the circle.

At the optimum, moving in that feasible direction cannot increase the objective to first order. Therefore

$$\nabla f(x^*)$$

must be orthogonal to the tangent direction.

But

$$\nabla g(x^*)$$

is also orthogonal to that tangent direction. In two dimensions, two nonzero vectors orthogonal to the same line must be parallel. Hence

$$\nabla f(x^*) = \lambda^* \nabla g(x^*).$$